UNDERBOT r07 — NOCO + T12 / REVIEW DRAFT

Approved power sources; one Adeept arm baseline; Kinect v2. No new 6S pack.

01 — NOCO input and distribution PCB boundary

File: 01_Power.kicad_sch

02 — Converters, stop interfaces and power cables

File: 02_Conversion.kicad_sch

03 — System interfaces and sensor harnesses

File: 03_System.kicad_sch

04 — Independent T12 base and integration holds

File: 04_Base.kicad_sch

Port-level module interconnection draft. HOLD labels are unresolved, not approved wiring.

See overall.svg, parts.xlsx and README.md. PCB layout/fabrication not released.

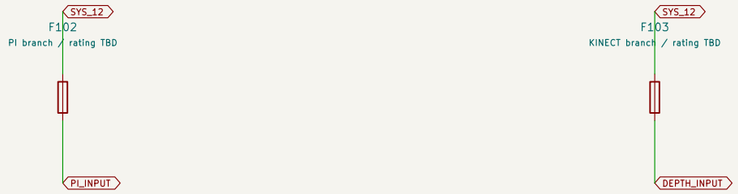
Sheet: /		
File: UnderBot.kicad_sch		
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KiCad E.D.A. 10.0.6		Id: 1/5

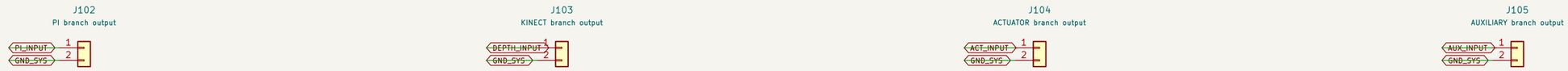
NOCO GB70 + T12 OEM BATTERY | r07 DESIGN REVIEW

Use GB70 12V OUT accessory port and approved accessory cable. Never use jump-start clamps or override.



Proposed PCB: input/output terminals, branch fuses, test points and ground distribution. Converter modules may be off-board.





NOCO nominal output: 12V / up to 15A. This is NOT a chosen fuse rating; branch currents and transient tests determine protection.

Input reverse-polarity / transient protection: DESIGN HOLD. Select after measuring GB70 output range and source shutdown.

NO CHARGING THROUGH THIS PCB. Charge NOCO separately using its specified input. Do not connect a 25.2V charger.

STOP HOLD: actuator branch requires hardwired manual-reset interruption. OEM T12 drive stop is separate and unverified.

This sheet specifies distribution interfaces. No PCB footprints, copper widths or fabrication files have been released.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN - NOT FOR FABRICATION

UnderBot

Sheet: /01 - NOCO input and distribution PCB boundary/
File: 01_Power.kicad_sch

Title: 01 - NOCO input and distribution PCB boundary

Size: A3	Date: 2026-10-02	**Rev: 0.7 NOCO REVIEW**
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COMMERCIAL POWER MODULES | PORT-LEVEL CONNECTIONS

9-24V DC input -> USB-C PD 5.15V / 5A.
Use supplied USB-C cable to Pi 5.
Vendor pages disagree on barrel dimensions:
verify actual connector before ordering harness.

A201
52Pi EP-0225 / DC input ONLY



A203
Hardwired stop + reset / HOLD



Kinect v2 requires its matching PC adapter.
Do not assume GB70 is a precision 12V regulator.
Direct protected feed only after output qualification;
otherwise select 12V buck-boost sized to actual label.

A202
12V conditioning / qualification HOLD



A204
VIN feed / verify documented 6-24V board



Documented controller VIN accepts 6-24V.
Confirm purchased revision before 12V VIN feed.
USB VBUS blocking is mandatory pending verification.

Auxiliary: speaker amplifiers / pan-tilt servos.
Pan-tilt motor power must also be interrupted on stop;
its final switching circuit is not released.
Stop/reset must prevent automatic restart and account
for arm collapse when torque is removed.

A205
5V auxiliary / current selection HOLD



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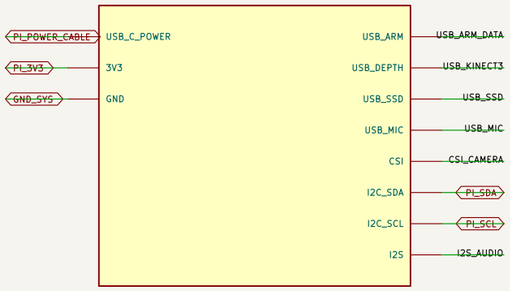
Sheet: /02 — Converters, stop interfaces and power cables/
File: 02_Conversion.kicad_sch

Title: 02 — Converters, stop interfaces and power cables

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SIGNAL CONNECTIONS | WHOLE ROBOT

A301
Existing Raspberry PI 5



A302
Adeept ADA031 kit incl. 6 servos



Use kit internal servo mapping. USB power backfeed
and emergency-stop behaviour remain qualification holds.

A303
Kinect v2 + matching PC adapter



A306
USB VBUS BLOCK / data preserved / HOLD

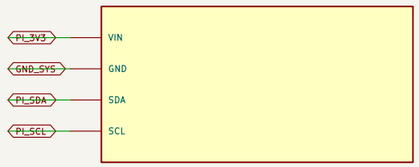


A304
BODY ICM20948 #4554



Address 0x68; strap AD0 per breakout guide.

A305
CAMERA ICM20948 #4554



Address 0x69; strap AD0 per breakout guide.

Pi pin mapping: 1=3V3; 3=GPIO2/SDA; 5=GPIO3/SCL; 6=GND. IMU interrupt/reset lines unused in this draft.

Additional interfaces: SSD + microphone via USB; Pi camera via correct 22-to-15 pin CSI cable; cooler via fan header.

4 ToF sensors require XSHUT sequencing or an I2C mux. Audio I2S and pan/tilt PWM pin assignment still pending.

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PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION
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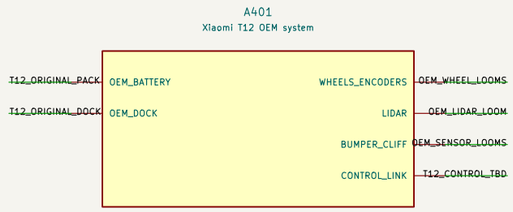
Sheet: /03 — System interfaces and sensor harnesses/
File: 03_System.kicad_sch

Title: 03 — System interfaces and sensor harnesses

Size: A3	Date: 2026-10-02	**Rev: 0.7 NOCO REVIEW**
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T12 RETAINS OEM BATTERY + CHARGING SYSTEM

Original battery: 14.8V; 3.2Ah nominal / 3.0Ah rated.
Keep OEM connectors and charging electronics.
T12 positive power rails never join GB70 positive rails.
Control/LiDAR protocol and signal-level interface are
not verified; no speculative Pi-to-OEM pin wiring.



BASE DRIVE STOP IS A RELEASE HOLD. Stopping the arm branch does not stop independently powered wheels.

Test OEM command access, navigation data access, added payload and tipping stability before free-moving operation.

Vacuum, brushes and mop pump assumed OFF for load estimate; independent control of these functions is unverified.

Current prototype baseline: ONE Adeept kit. A second kit is optional after load, mounting and firmware qualification.

J pin numbers are proposed harness assignments, not OEM connector pinouts
PROVISIONAL CIRCUIT / HARNESS DESIGN — NOT FOR FABRICATION

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Sheet: /04 — Independent T12 base and integration holds/
File: 04_Base.kicad_sch

Title: 04 — Independent T12 base and integration holds

Size: A3	Date: 2026-10-02	**Rev: 0.7 NOCO REVIEW**
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